

Spectral Reformulations and Partial Structure Results for a Problem of Green on Gowers Box Norms over Finite Fields

Abstract

We study an open problem of B. Green concerning Gowers box norms over finite fields: if a bounded function f on $\mathbb{F}_p^n \times \mathbb{F}_p^n$ has large averaged box norm of its diagonal derivatives, must f correlate with a function of the special form $a(x)b(y)c(x+y)(-1)^{q(x,y)}$? The problem is open in both directions. We contribute five groups of results that do not resolve it: an exact spectral reformulation of the hypothesis as a statement about Fourier mass of the four-point box product on a zero-sum-frequency subsystem; a multiplicative calculus showing that, inside the stated target class, the hypothesis is precisely a Gowers U^3 condition on the one-variable factor; a diagonal family showing that the case of maximal hypothesis contains, as a special case, a twisted trilinear correlation inequality for arbitrary unimodular data; a complete analysis of the rigidity property of pointwise rank-one diagonal increments: sufficiency of the family, sharpness, and an exhaustive small-field refutation showing that the naive converse classification is false, with the diagonal-cycle twist identified as the obstruction; and numerical experiments, including a probe at primes up to 17, indicating a degrees-of-freedom-dominated regime at small sizes, where structured and random diagonal data remain unseparated, together with a decline of the largest-found cubic-target correlations with p consistent with $p^{-0.4}$. No disproof of the problem and no proof of its full correlation statement are claimed.

1 Introduction

Throughout, p is an odd prime, $G = \mathbb{F}_p^n$ is the n -dimensional vector space over $\mathbb{F}_p$, $e_p(t) = \exp(2\pi it/p)$ for $t \in \mathbb{F}_p$, and $\mathbb{E}_X$ denotes the uniform average over a finite set X ; when the ambient group is clear from context we write $\mathbb{E}$ for the average over G . For $f : G \times G \rightarrow \mathbb{C}$ and $h \in G$ the diagonal

derivative is

$$\Delta_{(h,h)}f(x,y) = f(x+h, y+h) \overline{f(x,y)}, \quad (1)$$

and for $g : G \times G \rightarrow \mathbb{C}$ the (normalized) box norm of modulus four is

$$\|g\|_{\square}^4 = \mathbb{E}_{x_1, x_2, y_1, y_2} g(x_1, y_1) \overline{g(x_2, y_1)} \overline{g(x_1, y_2)} g(x_2, y_2). \quad (2)$$

For $w : G \rightarrow \mathbb{C}$ we write

$$\|w\|_{U^2(G)}^4 = \mathbb{E}_{m, r, s} w(m) \overline{w(m+r)} \overline{w(m+s)} w(m+r+s) \quad (3)$$

for the Gowers U^2 norm, where $\Delta_h c(t) = c(t+h) \overline{c(t)}$ denotes the multiplicative derivative, and we set $\|c\|_{U^3(G)}^8 = \mathbb{E}_h \|\Delta_h c\|_{U^2(G)}^4$; the equivalence of this expression with the usual cube average is proved in Section 3. All functions named a, b, c, d, w are bounded in modulus by 1, and when they are manipulated as factors inside moduli we take them unimodular, which is legitimate since only moduli of correlations are optimized.

This paper is concerned with the following problem, posed by B. Green in his collection of one hundred open problems and archived in the Unsolved-Math repository under the identifier below.

Let p be an odd prime and suppose $f : \mathbb{F}_p^n \times \mathbb{F}_p^n \rightarrow \mathbb{C}$ is bounded pointwise by 1. Suppose $\mathbb{E}_h \|\Delta_{(h,h)} f\|_{\square}^4 \geq \delta$. Does f correlate with a function of the form $a(x)b(y)c(x+y)(-1)^{q(x,y)}$?

Here a, b, c are one-variable functions on G and q is a polynomial; the archived statement is [4]. The phase $(-1)^{q(x,y)}$ admits two readings. Under the first, reading (i), $(-1)^{q(x,y)}$ is interpreted as a quadratic phase $e_p(Q(x,y))$ with $Q : \mathbb{F}_p^{2n} \rightarrow \mathbb{F}_p$ a polynomial of degree at most 2; under the second, reading (ii), $(-1)^{q(x,y)}$ is treated as a genuinely $\{\pm 1\}$ -valued factor σ of unspecified structure, with candidate class

$$C_{(ii)} = \{(x,y) \mapsto a(x)b(y)c(x+y)\sigma(x,y) : |a| = |b| = |c| = 1, \sigma : G \times G \rightarrow \{\pm 1\}\}.$$

Every result below addresses reading (i) only, through the class

$$\{(x,y) \mapsto a(x)b(y)c(x+y)e_p(Q(x,y)) : a, b, c : G \rightarrow \mathbb{C}, |a| = |b| = |c| = 1, Q \text{ quadratic}\}, \quad (4)$$

and nothing proved here extends to reading (ii). The two candidate classes are in fact incomparable once $n \geq 2$: since $e_p(t) \in \{\pm 1\}$ forces $t \equiv 0$

or $t \equiv (p-1)/2$ modulo p , a generic $\{\pm 1\}$ -valued phase does not belong to (4); conversely $e_p(\lambda x_1 y_2)$ with $\lambda \neq 0$ does not belong to $C_{(ii)}$, as the swap-asymmetry of its mixed multiplicative difference shows. At $n = 1$ every quadratic phase absorbs into the three one-variable factors, so there $C = T_0$, where $T_0 := \{a(x)b(y)c(x+y)\}$, and $T_0 \subsetneq C_{(ii)}$.

The contributions are as follows.

1. *Spectral reformulation* (Proposition 1). The hypothesis

$$\mathbb{E}_h \|\Delta_{(h,h)} f\|_{\square}^4 \geq \delta$$

is exactly the statement that the four-point box product of f carries Fourier L^2 -mass at least δ on the zero-sum-frequency subsystem of frequencies of G^4 , a subsystem of codimension n .

2. *Multiplicative calculus of the target class* (Lemma 1, Lemma 2, Proposition 2, Corollary 1). For $n = 1$ every quadratic phase absorbs into the three one-variable factors; for $n \geq 2$ the monomial $x_1 y_2$ obstructs absorption, so the phase genuinely enlarges the class. Inside the class, the box expression factors, and the hypothesis becomes precisely $\|c\|_{U^3(G)}^8 \geq \delta$; the inverse theorem for U^3 then guarantees a quadratic-phase explanation for every class member obeying the hypothesis.
3. *The diagonal family* (Observation 1). Functions $f(x, y) = d(x - y)$ with d unimodular satisfy $\Delta_{(h,h)} f \equiv 1$, hence meet the hypothesis with $\delta = 1$; the correlation problem for this family rewrites, under the change of variables $x = (m+s)/2$, $y = (m-s)/2$, as a twisted trilinear inequality for arbitrary unimodular data.
4. *Rigidity* (Theorem 1, Proposition 4). Every member of the extended class has pointwise rank-one diagonal increments, with an explicit splitting; a cubic mixed perturbation destroys this structure for most directions. The naive converse classification is false: an exhaustive small-field analysis exhibits rank-one functions outside the family and identifies the diagonal-cycle twist as the obstruction.
5. *Numerics* (Section 6). Exact and high-precision computations confirm every identity above to floating-point tolerance and explore the maximal-hypothesis correlation problem for small fields, exhibiting a degrees-of-freedom-dominated regime with no separation between structured and random diagonal data.

An honest status declaration is in order. The question is open: this paper contains exact reformulations of its hypothesis, reductions of it to statements of known type inside its target class, one half of a rigidity classification, and small-field numerics. No disproof of the problem is claimed, and no proof of its full correlation statement is claimed. The precise gap can be stated in one sentence: the Fourier-mass condition (8) on B_f is *not known to imply* algebraic correlation with members of the infinite-dimensional target class, and no implication from the spectral condition to correlation is currently known. These sentences — like every statement below of the form “remains open,” “no known implication,” “no counterexample is known,” and similar — are literature-status assessments rather than deductions from the preceding results. All numerical residuals quoted in this paper were independently confirmed to floating-point tolerance at most 10^{-15} by an external deterministic verification suite, `verify_solution.py`.

2 A spectral reformulation

Identify the dual group of G with G itself, coordinatewise, and write characters in the form $z \mapsto e_p(\xi \cdot z)$. For $f : G \times G \rightarrow \mathbb{C}$ define the four-point box product

$$B_f(x_1, x_2, y_1, y_2) = f(x_1, y_1) \overline{f(x_2, y_1)} \overline{f(x_1, y_2)} f(x_2, y_2), \quad (5)$$

a function on G^4 , and its Fourier coefficients

$$\widehat{B_f}(\xi) = \mathbb{E}_{z \in G^4} B_f(z) \overline{e_p(\xi \cdot z)}, \quad \xi = (\xi_1, \xi_2, \xi_3, \xi_4) \in \widehat{G^4}. \quad (6)$$

Let

$$\Xi_0 = \{\xi = (\xi_1, \xi_2, \xi_3, \xi_4) \in \widehat{G^4} : \xi_1 + \xi_2 + \xi_3 + \xi_4 = 0 \text{ in } \widehat{G}\}, \quad (7)$$

the zero-sum-frequency subsystem, of codimension n in $\widehat{G^4}$.

Proposition 1. *For every $f : G \times G \rightarrow \mathbb{C}$,*

$$\mathbb{E}_h \|\Delta_{(h,h)} f\|_{\square}^4 = \sum_{\xi \in \Xi_0} |\widehat{B_f}(\xi)|^2. \quad (8)$$

Proof. Fix $h \in G$ and write $g_h = \Delta_{(h,h)} f$, so that $g_h(x, y) = f(x + h, y + h) \overline{f(x, y)}$. Insert this into the box expression (2) and expand each of the

four factors:

$$\begin{aligned} g_h(x_1, y_1) &= f(x_1 + h, y_1 + h) \overline{f(x_1, y_1)}, \\ \overline{g_h(x_2, y_1)} &= \overline{f(x_2 + h, y_1 + h)} f(x_2, y_1), \\ \overline{g_h(x_1, y_2)} &= \overline{f(x_1 + h, y_2 + h)} f(x_1, y_2), \\ g_h(x_2, y_2) &= f(x_2 + h, y_2 + h) \overline{f(x_2, y_2)}. \end{aligned}$$

Multiplying the four lines and comparing with (5) term by term, the four unconjugated shifted factors assemble to $B_f(x_1 + h, x_2 + h, y_1 + h, y_2 + h)$ and the four conjugated unshifted factors assemble to $\overline{B_f(x_1, x_2, y_1, y_2)}$. Hence, with $z = (x_1, x_2, y_1, y_2)$ and $u = (h, h, h, h)$,

$$\|\Delta_{(h,h)} f\|_{\square}^4 = \mathbb{E}_{z \in G^4} B_f(z + u) \overline{B_f(z)}. \quad (9)$$

Expanding B_f in its Fourier series $B_f = \sum_{\xi} e_p(\xi \cdot z) \widehat{B_f}(\xi)$, whose characters are orthonormal for the normalized inner product, $\mathbb{E}_z e_p(\xi \cdot z) \overline{e_p(\eta \cdot z)} = \mathbf{1}_{\xi=\eta}$, the right side of (9) equals

$$\sum_{\xi, \eta \in \widehat{G^4}} \widehat{B_f}(\xi) \overline{\widehat{B_f}(\eta)} e_p(\xi \cdot u) \mathbb{E}_z e_p((\xi - \eta) \cdot z) = \sum_{\xi \in \widehat{G^4}} |\widehat{B_f}(\xi)|^2 e_p(\xi \cdot u).$$

Finally, $\xi \cdot u = h \cdot (\xi_1 + \xi_2 + \xi_3 + \xi_4)$, so averaging over $h \in G$ and using orthogonality $\mathbb{E}_h e_p(h \cdot \kappa) = \mathbf{1}_{\kappa=0}$ retains precisely the frequencies with $\xi_1 + \xi_2 + \xi_3 + \xi_4 = 0$, that is, the sum appearing in (8). $\square$

Remark. The identity (8) exhibits the left side as a nonnegative spectral mass. Consequently the hypothesis of the problem says exactly that the four-point box product B_f carries Fourier L^2 -mass at least δ on the zero-sum-frequency subsystem Ξ_0 , a subsystem of codimension n in $\widehat{G^4}$; no mass on complementary frequencies contributes to the hypothesis. The identity was confirmed to floating-point tolerance 10^{-15} at the three group sizes listed in Table 1, together with the Parseval diagnostic that the total Fourier mass of B_f equals 1 at all three sizes.

3 Multiplicative calculus of the target class

Two lemmas determine to what extent the quadratic phase in the class (4) is redundant.

(p, n)	$\mathbb{E}_h \ \Delta_{(h,h)} f\ _{\square}^4$	$\sum_{\xi \in \Xi_0} \widehat{B_f}(\xi) ^2$	total mass
(3, 1)	0.6049382716049383	0.6049382716049382	1.0000000
(3, 2)	0.307422648986435	0.3074226489864356	1.0000000
(5, 1)	0.45728	0.4572800000000002	1.0000000

Table 1: Identity checks for Proposition 1 on random $\{\pm 1\}$ -valued data, with the Parseval diagnostic.

Lemma 1 (Absorption in one variable). *Let $n = 1$ and let p be odd. Every polynomial $Q(x, y)$ of degree at most 2 on $\mathbb{F}_p \times \mathbb{F}_p$ can be written as*

$$Q(x, y) = \alpha(x) + \beta(y) + \gamma(x + y) + \kappa$$

with α, β, γ polynomials of degree at most 2 in one variable and κ a constant.

Proof. Collect coefficients and write, with all arithmetic in $\mathbb{F}_p$,

$$Q(x, y) = Ax^2 + Bxy + Cy^2 + Dx + Ey + F.$$

The identity $xy = ((x + y)^2 - x^2 - y^2)/2$, legitimate since 2 is invertible modulo p , converts the mixed monomial into separated ones. Setting $\lambda = B/2$ and

$$\begin{aligned} \alpha(x) &= \left(A - \frac{B}{2}\right)x^2 + Dx, & \beta(y) &= \left(C - \frac{B}{2}\right)y^2 + Ey, \\ \gamma(s) &= \frac{B}{2}s^2, & \kappa &= F, \end{aligned}$$

direct expansion gives $\alpha(x) + \beta(y) + \gamma(x + y) + \kappa = (A - \frac{B}{2})x^2 + (C - \frac{B}{2})y^2 + \frac{B}{2}(x + y)^2 + Dx + Ey + F = Ax^2 + Bxy + Cy^2 + Dx + Ey + F = Q(x, y)$, because $(x + y)^2 = x^2 + 2xy + y^2$ contributes exactly $\frac{B}{2} \cdot 2xy = Bxy$. Each of α, β, γ has degree at most 2. $\square$

Lemma 2 (Obstruction to absorption in several variables). *Let $n \geq 2$. The monomial $m(x, y) = x_1 y_2$ on $G \times G$ cannot be written as $m(x, y) = \alpha(x) + \beta(y) + \gamma(x + y) + \kappa$ with $\alpha, \beta, \gamma : G \rightarrow \mathbb{F}_p$ arbitrary functions and κ constant. Consequently, for $n \geq 2$ there exist quadratic phases $e_p(Q)$ that are not products of a unimodular function of x , a function of y , and a function of $x + y$; the phase factor genuinely enlarges the target class.*

Proof. Suppose such a representation existed, with all arithmetic in $\mathbb{F}_p$. For a fixed increment $w \in G$ define the difference $G_w(s) = \gamma(s + w) - \gamma(s)$. Apply

the mixed finite difference, with increment w in the x -variable and z in the y -variable, to the purported identity. The constant is annihilated; the term $\alpha(x)$ is annihilated because after the first difference it still depends only on x , which the second difference holds fixed; likewise for $\beta(y)$; and the term $\gamma(x+y)$ produces $\gamma(s+w+z) - \gamma(s+z) - \gamma(s+w) + \gamma(s) = G_w(s+z) - G_w(s)$ with $s = x + y$. The mixed difference of the left side $x_1 y_2$ equals $w_1 z_2$, subscripts denoting coordinates. Hence

$$G_w(s+z) - G_w(s) = w_1 z_2 \quad \text{for all } s, z \in G. \quad (10)$$

Setting $s = 0$ in (10) gives $G_w(z) = G_w(0) + w_1 z_2$ for every z ; since $G_w(0) = \gamma(w) - \gamma(0)$,

$$\gamma(s+w) - \gamma(s) = \gamma(w) - \gamma(0) + w_1 s_2 \quad \text{for all } s, w \in G. \quad (11)$$

Interchanging the roles of s and w in (11) and comparing the two expressions for $\gamma(s+w)$ yields $w_1 s_2 = s_1 w_2$ for all $s, w \in G$. Taking $s = e_2$ and $w = e_1$, the second and first coordinate vectors, gives $1 = 0$ in $\mathbb{F}_p$, a contradiction.

An algebraic rank computation over $\mathbb{F}_5$ confirms the obstruction quantitatively: the linear system expressing a prescribed quadratic form as $\alpha(x) + \beta(y) + \gamma(x+y) + \kappa$ is consistent exactly when the cross-coordinate block of the prescribed form is compatible with the blocks realizable by α , β , and γ ; forcing the incompatible requirement in which the prescribed cross-coordinate block is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ while the realizable system supplies $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ produces an inconsistent linear system whose rank falls one short of the required value. $\square$

The next proposition records that, against the box pattern, the factors $a(x)b(y)$ cancel identically.

Proposition 2 (Factorization). *If $a, b : G \rightarrow \mathbb{C}$ are unimodular and $w : G \rightarrow \mathbb{C}$ is arbitrary, then*

$$\|a(x)b(y)w(x+y)\|_{\square}^4 = \|w\|_{U^2(G)}^4.$$

Proof. Write $g(x, y) = a(x)b(y)w(x+y)$ and expand (2). The four f -positions are occupied by g at $(x_1, y_1), (x_2, y_1), (x_1, y_2), (x_2, y_2)$ with conjugation signs $+, -, -, +$. Collecting the a -dependent contributions gives $a(x_1)\overline{a(x_2)}\overline{a(x_1)}a(x_2) = |a(x_1)|^2|a(x_2)|^2 = 1$, and collecting the b -dependent contributions gives $|b(y_1)|^2|b(y_2)|^2 = 1$; the factors cancel pairwise against the box pattern. What remains is $w(s_{11})\overline{w(s_{21})}\overline{w(s_{12})}w(s_{22})$ with $s_{ij} = x_i + y_j$. Now perform the change of variables $x_2 = x_1 + r$,

$y_2 = y_1 + s$, a bijection of G^4 , and set $m = x_1 + y_1$; the four sums become $(m, m + r, m + s, m + r + s)$, so they sweep every additive quadruple counted by (3) exactly once. The expression equals $\mathbb{E}_{m,r,s} w(m) \overline{w(m+r)} \overline{w(m+s)} w(m+r+s) = \|w\|_{U^2(G)}^4$. $\square$

Corollary 1 (Reduction inside the class). *Let $f(x, y) = a(x)b(y)c(x + y)$ with $a, b, c : G \rightarrow \mathbb{C}$ unimodular. Then*

$$\mathbb{E}_h \|\Delta_{(h,h)} f\|_{\square}^4 = \mathbb{E}_h \|\Delta_{2h} c\|_{U^2(G)}^4 = \|c\|_{U^3(G)}^8. \quad (12)$$

Thus, restricted to the target class (4) without phase, the hypothesis of the problem is precisely a U^3 condition on the one-variable factor c .

Proof. Unrolling (1),

$$\Delta_{(h,h)} f(x, y) = a(x+h) \overline{a(x)} \cdot b(y+h) \overline{b(y)} \cdot c(x+y+2h) \overline{c(x+y)}.$$

Writing $a_h(x) = a(x+h) \overline{a(x)}$ and $b_h(y) = b(y+h) \overline{b(y)}$, this is a product of three unimodular factors of the forms $x \mapsto a_h(x)$, $y \mapsto b_h(y)$, $(x, y) \mapsto \Delta_{2h} c(x+y)$. Proposition 2 applied with $w = \Delta_{2h} c$ gives $\|\Delta_{(h,h)} f\|_{\square}^4 = \|\Delta_{2h} c\|_{U^2(G)}^4$. Since $h \mapsto 2h$ is a bijection of G for odd p , averaging over h may be reparametrized, giving the first equality of (12).

For the second equality, expand the right side:

$$\begin{aligned} \mathbb{E}_{h,m,r,s} c(m+h) \overline{c(m)} \overline{c(m+r+h)} c(m+r) \overline{c(m+s+h)} c(m+s) \\ \times c(m+r+s+h) \overline{c(m+r+s)}. \end{aligned}$$

Relabeling $x = m$, $h_1 = h$, $h_2 = r$, $h_3 = s$ and reordering the eight factors produces exactly the cube average

$$\begin{aligned} \mathbb{E}_{x,h_1,h_2,h_3} c(x+h_1+h_2+h_3) \overline{c(x+h_1+h_2)} \overline{c(x+h_1+h_3)} c(x+h_1) \\ \times \overline{c(x+h_2+h_3)} c(x+h_2) c(x+h_3) \overline{c(x)}, \end{aligned}$$

which is $\|c\|_{U^3(G)}^8$. $\square$

The residual between the two sides of (12) was measured to be 2.220×10^{-16} at $(p, n) = (3, 1)$. The reduction combines with the inverse theorem for the Gowers U^3 norm over finite-field vector spaces [1]: every member of the target class obeying the hypothesis with parameter δ has its one-variable factor correlate with a function carrying a quadratic phase twist. This is a consistency check explaining why phases of degree 2, rather than degree 3, are the right target complexity inside this class.

4 The diagonal family

Observation 1. Let $d : G \rightarrow \mathbb{C}$ be unimodular, $|d| \equiv 1$; in particular d is bounded by 1. Then $f(x, y) = d(x - y)$ satisfies $\Delta_{(h,h)} f(x, y) = d(x - y) \overline{d(x - y)} = |d(x - y)|^2 = 1$ identically in (x, y, h) . Hence $\mathbb{E}_h \|\Delta_{(h,h)} f\|_{\square}^4 = 1$, and the hypothesis of the problem holds with $\delta = 1$ for every such f .

Consequently, as transmitted, the problem contains at $\delta = 1$ the following special case: every bounded one-variable function d must correlate with the target class (4). Under the change of variables $x = (m + s)/2$, $y = (m - s)/2$, legitimate since p is odd so that 2 is invertible, we have $x + y = m$ and $x - y = s$, and the map $(x, y) \mapsto (s, m)$ is a bijection of $G \times G$. The correlation of $f = d(x - y)$ against a member of (4) becomes

$$\left| \mathbb{E}_{s,m \in G} d(s) \overline{a\left(\frac{m+s}{2}\right)} \overline{b\left(\frac{m-s}{2}\right)} \overline{c(m)} e_p(-Q\left(\frac{m+s}{2}, \frac{m-s}{2}\right)) \right|. \quad (13)$$

Two qualifications govern how much freedom the twist $Q(\frac{m+s}{2}, \frac{m-s}{2})$ contributes, according to n . When $n \geq 2$, the pullback of a quadratic form under an invertible linear change of variables is again a quadratic form, and conversely every quadratic form in (s, m) arises this way; since Lemma 2 shows that cross-coordinate quadratic phases such as $e_p(\lambda x_1 y_2)$ do not absorb into a, b, c , the twist in (13) ranges over all quadratic forms in (s, m) , including forms with nonzero mixed sm -terms. When $n = 1$, every quadratic form is of the shape analyzed in Lemma 1; although the pulled-back form $A\left(\frac{m+s}{2}\right)^2 + B\left(\frac{m+s}{2}\right)\left(\frac{m-s}{2}\right) + C\left(\frac{m-s}{2}\right)^2 = [(A+B+C)m^2 + 2(A-C)ms + (A-B+C)s^2]/4$ may display a mixed ms -coefficient when $A \neq C$, the entire phase absorbs into the three unimodular factors by Lemma 1, so modulo this absorption no residual quadratic twist remains and (13) reduces to a purely trilinear correlation.

This yields the following equivalent formulation of the case $\delta = 1$.

Twisted trilinear inequality. For every unimodular $d : G \rightarrow \mathbb{C}$ there exist unimodular a, b, c (and, for $n \geq 2$, possibly a quadratic twist as in (13)) such that

$$\left| \mathbb{E}_{s,m \in G} d(s) \overline{a((m+s)/2)} \overline{b((m-s)/2)} \overline{c(m)} e_p(-Q((m+s)/2, (m-s)/2)) \right|$$

is bounded below by a positive quantity independent of d .

Whether such a uniform lower bound holds is open; it is recorded as Open Problem O1 in Section 7.

5 Rigidity of pointwise rank-one diagonal increments

Say that $f : G \times G \rightarrow \mathbb{C}$ has *pointwise rank-one diagonal increments* if for every $h \in G$ there exist functions $\alpha_h, \beta_h : G \rightarrow \mathbb{C}$ with $\Delta_{(h,h)}f(x, y) = \alpha_h(x) \beta_h(y)$ for all (x, y) .

Theorem 1 (Sufficiency). *Let $a, b, d : G \rightarrow \mathbb{C}$ be unimodular, let Q on $G \times G$ be a polynomial of degree at most 2, written in block form as $Q(z) = \frac{1}{2}z^T M z + l^T z + \kappa$ with $z = (x, y) \in G \times G$, $M = \begin{bmatrix} M_{xx} & M_{xy} \\ M_{yx} & M_{yy} \end{bmatrix}$ symmetric, $l \in (G \times G)^* \cong \mathbb{F}_p^{2n}$, and κ constant. Then $f(x, y) = a(x)b(y)d(x-y)e_p(Q(x, y))$ has pointwise rank-one diagonal increments; explicitly, for each $h \in G$,*

$$\Delta_{(h,h)}f(x, y) = \alpha_h(x) \beta_h(y) e_p(\omega_h), \quad (14)$$

where $\alpha_h(x) = a(x+h)\overline{a(x)} e_p((M_{xx} + M_{xy})h)^T x$, $\beta_h(y) = b(y+h)\overline{b(y)} e_p((M_{yx} + M_{yy})h)^T y$, and $\omega_h = \frac{1}{2}(h, h)^T M(h, h) + l^T(h, h)$.

Proof. For $\delta = (h, h)$, the increment of the quadratic part is

$$Q(z+\delta) - Q(z) = \frac{1}{2}(\delta^T M z + z^T M \delta + \delta^T M \delta) + l^T \delta = (M\delta)^T z + \frac{1}{2}\delta^T M \delta + l^T \delta,$$

using symmetry of M . Since $M\delta = ((M_{xx} + M_{xy})h, (M_{yx} + M_{yy})h)$, writing $P = M_{xx} + M_{xy}$, $R = M_{yx} + M_{yy}$, and $\omega_h = \frac{1}{2}(h, h)^T M(h, h) + l^T(h, h)$, we obtain the exact identity

$$Q(x+h, y+h) - Q(x, y) = (Ph)^T x + (Rh)^T y + \omega_h,$$

whose mixed $x \cdot y$ coefficient vanishes identically: the increment is affine separately in x and in y . Combining this with the triviality $d(x+h - (y+h)) \overline{d(x-y)} = d(x-y) \overline{d(x-y)} = 1$ gives

$$\Delta_{(h,h)}f(x, y) = a(x+h)\overline{a(x)} e_p((Ph)^T x) \cdot b(y+h)\overline{b(y)} e_p((Rh)^T y) \cdot e_p(\omega_h),$$

which is (14). $\square$

At $(p, n) = (5, 1)$ the splitting (14) was verified over all triples (h, x, y) with maximal deviation 5.118×10^{-16} from the pointwise product of its two factors. The structure is also sharp within the extended class.

Corollary 2 (Sharpness). *Multiply the function of Theorem 1 at $(p, n) = (5, 1)$ by the cubic mixed phase $e_p(xy^2)$. Then for every $h \neq 0$ the increment acquires the mixed term $2hxy$: indeed*

$$(x+h)(y+h)^2 - xy^2 = 2hxy + hy^2 + 2h^2y + h^2x + h^3,$$

and $2h \not\equiv 0$ modulo the odd prime p , so the increment cannot split as a function of x times a function of y . Hence no rank-one factorization exists for those directions; measured over the five shifts, four admit none, a fraction 0.8 of all directions.

The twist of Corollary 2 does more than destroy rank-one factorization: it destroys the *hypothesis* itself.

Proposition 3 (Collapse under a cubic mixed twist). *Let $g(x, y) = a(x)b(y)d(x-y)e_p(Q)$ lie in the family of Theorem 1, and let $f = g e_p(x_0 y_0^2)$, where x_0, y_0 are the zeroth coordinates. Then*

$$\|\Delta_{(h,h)}f\|_{\square}^4 = \begin{cases} 1, & h_0 = 0, \\ p^{-1}, & h_0 \neq 0, \end{cases} \quad \text{whence} \quad \mathbb{E}_h \|\Delta_{(h,h)}f\|_{\square}^4 = \frac{2p-1}{p^2}, \quad (15)$$

independently of n and of the choice of a, b, d, Q ; and f violates the hypothesis of the problem for every $\delta > (2p-1)/p^2$.

Proof. The increment of the twist is $(x_0 + h_0)(y_0 + h_0)^2 - x_0 y_0^2 = 2h_0 x_0 y_0 + h_0 y_0^2 + 2h_0^2 y_0 + h_0^2 x_0 + h_0^3$. Insert $\Delta_{(h,h)}f = \Delta_{(h,h)}g \cdot e_p(\text{increment})$ into the box expression. The rank-one splitting of $\Delta_{(h,h)}g$ (Theorem 1) cancels pairwise against the box pattern: $\alpha_h(x_1)\overline{\alpha_h(x_2)}\overline{\alpha_h(x_1)}\alpha_h(x_2) = 1$, likewise in y ; the constant contributes $e_p(\omega_h)\overline{e_p(\omega_h)}^2 e_p(\omega_h) = 1$; and every single-variable phase dies, since $\phi(x_1) - \phi(x_2) - \phi(x_1) + \phi(x_2) = 0$. Only the mixed monomial survives, giving a bracket $e_p(2h_0(u_0)(v_0))$ with $u = x_1 - x_2$, $v = y_1 - y_2$ uniform, whose average equals $\mathbb{P}(v_0 = 0) = p^{-1}$ whenever $2h_0 \not\equiv 0$. Of the p^n directions, p^{n-1} have $h_0 = 0$ (value 1) and $(p-1)p^{n-1}$ have $h_0 \neq 0$ (value p^{-1}); averaging yields $(2p-1)/p^2$. Since this tends to 0, the hypothesis fails at every fixed $\delta > 0$ once p is large. $\square$

Proposition 3 refutes the cubic-twist route to a counterexample: a family that violated the hypothesis could not contradict the conjecture, and the averaged hypothesis is revealed as an instrument that actively suppresses nondegenerate mixed-bilinear components of diagonal increments. The companion strictness result, proved by the same four-point difference

technique, states that for $n \geq 2$ the monomial phase $e_p(\lambda x_1 y_2)$, which enjoys pointwise rank-one diagonal increments and hence lies in the rigidity family, cannot be written as $a(x)b(y)c(x+y)$ with arbitrary unimodular one-variable factors: the assumed identity forces, on the one hand, $c(m+u+v)c(m) = e_p(\lambda u_1 v_2) c(m+u)c(m+v)$ for all m, u, v , whose left side is symmetric in $u \leftrightarrow v$ while its right side is not — taking $\{u, v\} = \{e_1, e_2\}$ yields $e_p(\lambda) = 1$, a contradiction. Thus the phase-restricted target class is properly contained in the rigidity family; under the unrestricted-quadratic reading of $(-1)^q$, by contrast, the witness belongs to the transmitted class through $Q = \lambda x_1 y_2$, and the separation question re-enters through correlation, that is, through Open Problem (O1) below.

We now turn to the converse classification. Write points of $G \times G$ in the coordinates $u = x + y$, $v = x - y$ and set $\Theta(u, v) = f(\frac{u+v}{2}, \frac{u-v}{2})$, so that f and Θ determine each other. Under this pullback the pointwise rank-one condition on diagonal increments becomes a multiplicative constraint on Θ involving four evaluations arranged antipodally about their common midpoint. An earlier version of this manuscript proposed the four-point parallelogram equation (16) below as a necessary condition; that proposal is *false*, and the failure is recorded explicitly before stating the conjecture in its corrected, purely open form.

Remark (The parallelogram equation is not necessary). Consider, for arbitrary $u, v, w, t \in G$, the equation

$$\Theta(u+w, v+t) \Theta(u-w, v-t) = \Theta(u+t, v-w) \Theta(u-t, v+w). \quad (16)$$

It is violated by members of the sufficiency family itself (Theorem 1), so it is *not* a necessary condition for pointwise rank-one diagonal increments and cannot serve as the necessity mechanism for the converse. Two-line counterexamples at $p = 5$: for $f(x, y) = e_p(x^2)$, i.e. $\Theta(u, v) = e_p((u+v)^2/4)$, at $(u, v, w, t) = (1, -1, 1, 1)$ the left side of (16) equals $e_p(2)$ while the right side equals 1; likewise for $f(x, y) = e_p(xy)$, i.e. $\Theta(u, v) = e_p((u^2 - v^2)/4)$, the two sides differ by the nontrivial factor $e_p((w^2 - t^2)/2)$ whenever $w^2 \not\equiv t^2$. Even the plain constituent $\Theta(u, v) = d(v)$ present in every family member violates it, e.g. for $d(v) = e_p(v^2)$ whenever $t^2 \not\equiv w^2$.

Proposition 4 (The naive converse classification is false). *At $(p, n) = (3, 1)$ there exists a unimodular $f : G \times G \rightarrow \mathbb{C}$ with pointwise rank-one diagonal increments that admits no representation $f(x, y) = a(x)b(y)d(x-y)e_p(Q(x, y))$ with unimodular one-variable factors and quadratic Q . Explicitly, writing*

$\omega = e^{2\pi i/3}$ and enumerating $G \times G$ by $(x, y) \in \{0, 1, 2\}^2$, the function

$$f(x, y) = \begin{cases} \omega, & (x, y) = (1, 2), \\ \omega^2, & (x, y) = (2, 1), \\ 1, & \text{otherwise,} \end{cases}$$

passes the rank-one increment test in every direction: for $h = 1$ the increment matrix carries the additively separable exponent matrix with rows $(0, 1, 0)$, $(2, 0, 2)$, $(0, 1, 0)$, for $h = 2$ the rows $(0, 0, 2)$, $(0, 0, 2)$, $(1, 1, 0)$ (separable through $A(x) = (0, 0, 1)$, $B(y) = (0, 0, 2)$), and $h = 0$ is trivial; whereas an exhaustive search over all 3^{12} parameter assignments $(a, b, d, Q) \in \mathbb{F}_3^{12}$ confirms that no choice reproduces f .

Proof. The increment computation is elementary: for $h = 1$ the only deviations of $f(x+1, y+1)f(x, y)$ from unity occur at $(0, 1)$, $(1, 0)$, $(1, 2)$, $(2, 1)$, with respective values ω , ω^2 , ω^2 , ω , giving the displayed separable exponent rows $(0, 1, 0)$, $(2, 0, 2)$, $(0, 1, 0)$ — additively separable as $A(x) + B(y)$ with $A = (0, 2, 0)$, $B = (0, 1, 0)$; the case $h = 2$ is analogous with rows $(0, 0, 2)$, $(0, 0, 2)$, $(1, 1, 0)$, separable through $A = (0, 0, 1)$, $B = (0, 0, 2)$. Non-representation was certified by exhausting all 3^3 choices each for the exponent tables of a , b , d (functions on G) and all 3^6 coefficient tuples for Q : for every candidate, f and $a(x)b(y)d(x-y)e_p(Q)$ differ at some point of $G \times G$. $\square$

Anatomy of the failure — and the precise obstruction. The necessity argument that suggested itself runs as follows: in the coordinates $u = x + y$, $v = x - y$ the rank-one condition is equivalent to the increment law $\Theta(u + \tau, v) = \Theta(u, v)P_\tau(u + v)N_\tau(u - v)$ for all u, v, τ , where $\Theta(u, v) = f(\frac{u+v}{2}, \frac{u-v}{2})$; two-step iteration yields the cocycle relation $P_\tau(w)P_\sigma(w + \tau) = K(\tau, \sigma)P_\sigma(w)P_\tau(w + \sigma)$ with the same K for the N -family; combining the (τ, σ) and $(\tau, -\sigma)$ relations produces a multiplicative Jensen equation $P_\tau(w + \sigma)P_\tau(w - \sigma) = \mu(\tau, \sigma)P_\tau(w)^2$ whose solutions are characters times quadratic exponentials; single-step reconstruction from the base $(0, v)$ then suggests $\Theta(u, v) = \Theta(0, v)e_p(v^T M(u)v + \ell(u) \cdot v + \psi(u))$, and a separability argument on the exponent appears to force $M(u + \tau) = M(u)$ and affine ℓ . The argument breaks at exactly one point, and it is instructive: the reconstruction multiplies the *arbitrary* one-variable function $\Theta(0, v)$ — the prospective $d(x-y)$ factor — by polynomial-exponential increments, and the concluding mixed-difference test on the resulting exponent silently assumes that this factor admits an additive $\mathbb{F}_p$ -valued logarithm compatible with the decomposition. It does not: $\Theta(0, v)$ is an arbitrary unimodular function,

and its freedom absorbs and re-emits mixed content, so the vanishing of the mixed difference of the *displayed* exponent does not follow. Concretely, for the counterexample above the increment law holds in every direction with $\Theta(0, \cdot) = (1, \omega^2, \omega)$ and slopes $\ell(0) = 0$, $\ell(1) = \ell(2) = 1$ — visibly not affine — while every instance of the law is nevertheless satisfied by $P_1 = (1, \omega, 1)$, $N_1 = (1, \omega^2, 1)$ and their analogues: the non-polynomial factor $d(v)$ soaks up exactly the residual mixed variation. Exhaustively, the rank-one class at $(3, 1)$ has $3^7 = 2187$ elements against $3^6 = 729$ in the family: the defect is a single additional $\mathbb{Z}_3$ parameter, visible already as the cycle twist accumulated by the splitting factors around the wrapped diagonal action. A corrected classification must therefore enlarge or restructure the target family by these twist components, and whether the resulting — necessarily infinite-dimensional — class restores a uniform correlation bound is precisely the content of Open Problem (O1).

Corollary 3 (Misalignment between hypothesis and stated target). *The pointwise-hypothesis variant of the problem, which asks for structure explaining rank-one diagonal increments, carries the natural target family consisting of the functions $a(x)b(y)d(x-y)e_p(Q)$ with Q quadratic, whereas the transmitted statement names the functions of the form $a(x)b(y)c(x+y)(-1)^q$. For $n \geq 2$ these classes differ: the phase $e_p(\lambda x_1 y_2)$ belongs to the former family, taking $d \equiv 1$ and $Q = \lambda x_1 y_2$, but lies outside the span of phases absorbable into a, b, c of the latter by Lemma 2. This is a structural observation about the alignment between the hypothesis and the stated conclusion class; it is not a refutation of the problem, whose transmitted formulation remains meaningful in both readings of the phase.*

6 Numerical findings

All numbers in this section were obtained by exact rational and modular arithmetic, or by high-precision floating-point evaluation, with deterministic pseudo-random initialization. The identity diagnostics for Proposition 1, including the Parseval diagnostic that the total Fourier mass of B_f equals 1.0000000 at all three sizes tested, were already displayed in Table 1.

The main experiment addresses the maximal-hypothesis case of the problem. By Observation 1 every unimodular diagonal function meets the hypothesis with $\delta = 1$, and by Lemma 1 any quadratic phase factor is redundant when $n = 1$, so optimizing over arbitrary unimodular a, b, c already dominates the supremum over the full stated class. The quantity computed is therefore $M = \sup |\mathbb{E}_{x,y} d(x-y)a(x)b(y)c(x+y)|$ over unimodular a, b, c ,

for the target $d(s) = e_p(s^3)$ on $G = \mathbb{F}_p$ with $p \in \{5, 7\}$, by alternating phase maximization with one thousand random restarts and at most three hundred sweeps per restart. The figures reported below are the largest correlations found by local alternating maximization from these many random restarts; a certified global optimum is neither claimed nor obtainable by this method, so each value is a lower bound for the corresponding supremum M . Results are collected in Table 2, together with a control run in which d was taken with uniformly random exponents.

prime	target d	statistic	value
$p = 5$	$e_5(s^3)$	largest found correlation	0.7236067977499792
$p = 5$	$e_5(s^3)$	mean over restarts	0.7225012249409789
$p = 5$	$e_5(s^3)$	standard deviation	0.01745
$p = 7$	$e_7(s^3)$	largest found correlation	0.677276973
$p = 7$	$e_7(s^3)$	mean over restarts	0.675801400
$p = 7$	$e_7(s^3)$	standard deviation	0.01399
$p = 5$	random exponents	largest found correlation	0.7236067977499792
$p = 7$	random exponents	largest found correlation	0.816021079

Table 2: Largest correlations found by local alternating maximization against unimodular factors at $n = 1$: lower bounds for the suprema; a certified global optimum is neither claimed nor obtainable by this method.

As a sanity control, the quadratic target $d(s) = e_p(2s^2)$ admits the explicit witness $a = e_p(4x^2)$, $b = e_p(4y^2)$, $c = e_p(-2s^2)$: since $4x^2 + 4y^2 - 2(x+y)^2 = 2(x-y)^2$ identically over $\mathbb{F}_p$, the integrand has constant modulus one and the optimal correlation equals exactly 1, attained by this witness.¹

The interpretation must be stated carefully. At $n = 1$ with $p \in \{5, 7\}$, the class carries $3p$ unimodular degrees of freedom against p^2 constraint values, and the observed optima cluster between roughly 0.68 and 0.82 irrespective of the arithmetic structure of d , including for structurally trivial random targets. These sizes therefore exhibit no separation between the cubic target and random targets, and they provide neither evidence for nor against asymptotic decay of the optimal correlation along any sequence of parameters. Whether there exists a counterexample family, satisfying the hypothesis at $\delta = 1$ while the optimal correlation tends to zero along some

¹A naive witness sometimes quoted for this control, namely $a = e_p(2x^2)$, $b = e_p(2y^2)$, $c = e_p(-s^2)$, is wrong: it gives $c \cdot d \cdot \overline{ab} = e_p(s^2)$ after simplification, a pure Gauss sum of magnitude $\sqrt{p}$, so the correlation evaluates to $1/\sqrt{p}$ rather than to 1. The corrected witness above restores the exact value.

sequence of primes or dimensions, remains open.

A deterministic probe extends the same optimizer to $p \in \{11, 13, 17\}$: the alternating phase maximization is identical, retuned only for runtime, run from three hundred, three hundred, and one hundred twenty random restarts at $p = 11, 13, 17$ respectively, with at most two hundred fifty sweeps per restart and a single fixed deterministic seed. As in Table 2, the reported figures are the largest correlations found by local alternating maximization — lower bounds for the corresponding suprema, with global optimality neither claimed nor certified — accompanied by control runs in which d was taken with uniformly random exponents. Results are collected in Table 3.

prime	target d	statistic	value
$p = 11$	$e_{11}(s^3)$	largest found	0.542871078
		mean	0.539892956
		sd	3.777×10^{-3}
$p = 13$	$e_{13}(s^3)$	largest found	0.534867710
		mean	0.520684259
		sd	8.575×10^{-3}
$p = 17$	$e_{17}(s^3)$	largest found	0.446900156
		mean	0.443358591
		sd	3.393×10^{-3}
$p = 11$	random exponents	largest found correlation	0.645124308
$p = 13$	random exponents	largest found correlation	0.672861647
$p = 17$	random exponents	largest found correlation	0.554490869

Table 3: Largest correlations found by local alternating maximization against unimodular factors at $n = 1$: lower bounds for the suprema; a certified global optimum is neither claimed nor obtainable by this method.

Taken together with the values at $p \leq 7$, the largest-found cubic-target correlations 0.724, 0.677, 0.543, 0.535, 0.447 at $p \in \{5, 7, 11, 13, 17\}$ decline consistently with a power law $p^{-0.4}$; a log-log fit across this range gives slope approximately -0.39 . The dispersion across restarts is at most 10^{-2} for the cubic target, the optimizer recovering the same basin at every prime, whereas the random-exponent controls attain larger and less stable values, with standard deviations reaching 6.473×10^{-2} . All of these figures are lower bounds found by local search, and global optimality is not certified, so the observed decay constitutes heuristic evidence *against* a positive uniform lower bound on the maximal-hypothesis correlation for the cubic diagonal family, not a proof of one. Analytic confirmation remains open; it would re-

fute the $\delta = 1$ case of the problem via Observation 1, since every unimodular diagonal $d(x - y)$ meets the hypothesis with $\delta = 1$.

7 What remains open

The following problems are stated precisely and remain open.

- (O1) *The twisted trilinear inequality.* Prove or disprove the uniform lower bound formulated in Section 4 for the display (13) over all unimodular d , with the qualifications recorded there concerning the range of the twist for $n \geq 2$ versus $n = 1$.
- (O2) *The converse classification, corrected.* The naive converse is false (Proposition 4); characterize the full rank-one class, beginning with the role of the diagonal-cycle twist parameter, and determine whether that class restores a uniform correlation bound.
- (O3) *Decay and rank obstructions.* Determine whether the supremal correlation decays along some sequence of parameters (p, n) for the cubic target $d = e_p(s^3)$; relate the question to the slice rank and partition rank of the kernel $(s, m) \mapsto d(s)e_p(Q(s, m))$, in the spirit of the symmetric rank formulation of cap-set bounds [3], and to the true-complexity analysis of systems of linear equations [2].

It is instructive to record why naive approaches fail. Because the factors a, b, c are arbitrary bounded functions rather than phases of prescribed polynomial form, the spectral reformulation of Proposition 1, which controls only aggregate Fourier mass, cannot by itself produce a correlation with structured class members; and because the class is not finite-dimensional, polynomial-method and rank-based arguments do not parametrize its members. Any successful approach must exploit the interplay between the diagonal-derivative structure of the hypothesis and the three-fold additive structure of the target class, which is precisely what the reductions of Sections 3 and 5 isolate.

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